American Put Pricing under the Lin–Lin–He Stochastic Volatility Model

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Contents

1 Introduction **2**

2 The LLH Model **2**

3 Pricing Approach **3**
3.1 Mathematical Formulation . . . 3
3.2 Estimation Framework . . . 4
3.2.1 LSM Algorithm . . . 4
3.2.2 Variance Reduction with Rasmussen Control Variates . . . 5
3.3 European Price of the Control Variate . . . 6
3.4 Confidence Interval Construction . . . 6
3.5 Computational Complexity . . . 7

4 American Put Pricing Results **8**
4.1 Experimental Setup . . . 8
4.2 Price vs Spot at $K = 100$ . . . 9
4.3 Price across Moneyness . . . 9
4.4 Convergence with Path Count . . . 11

5 Cost–Precision Tradeoff **12**
5.1 Variance Reduction Analysis . . . 12
5.2 Computational Cost . . . 13

6 Further Validation **15**
6.1 Black–Scholes Limit Validation . . . 15
6.2 Early Exercise Premium . . . 15

7 Conclusion **16**

1 Introduction

This report presents a numerical study of American put option pricing via the Longstaff–Schwartz least squares Monte Carlo algorithm (LSM) under the improved Stein–Stein stochastic volatility model proposed by Lin et al. (2024), hereafter referred to as the LLH model. The companion report on European pricing established the validity of the semi-analytical LLH formula; here we use the formula to compute a Rasmussen-style control variate to reduce the variance of the LSM estimator.

The central question is whether control variates remain effective under stochastic volatility, as they are under geometric Brownian motion (GBM). Given the semi-analytical nature of the LLH formula, another question is the computational stability and cost of using a Rasmussen-style control variate.

We investigate these questions by:

1. comparing three LSM price estimators – plain, with a BS control variate proxy (CV-BS), and with control variate using the LLH European formula (CV-LLH)– across moneyness and maturity scenarios;
2. validating CV-BS as a near-perfect control variate in the Black–Scholes limit;
3. quantifying variance reduction ratios across parameter regimes;
4. computing the early exercise premium (EEP) and verifying its consistency across methods.

All experiments are repeated under two reference parameter sets (Tables 1 and 2 from Lin et al. 2024) to assess robustness across calibration regimes.

2 The LLH Model

The LLH model specifies the following system of stochastic differential equations under the risk-neutral measure:

$$\frac{dS_t}{S_t} = r\,dt + \sigma_t\,dW_t^1, \tag{1}$$

$$d\sigma_t = \kappa(\theta_t - \sigma_t)\,dt + \nu\,dW_t^2, \tag{2}$$

$$d\theta_t = \lambda\,dt + \eta\,dB_t, \tag{3}$$

$$dB_t = B_t\,dW_t, \quad B_0 = 1, \tag{4}$$

where $\mathrm{Cov}(W_t^1, W_t^2) = \rho\,t$ and W_t is an independent standard Brownian motion. The model nests Black–Scholes (Black and Scholes, 1973) ($\kappa = \nu = \lambda = \eta = \rho = 0$), Stein–Stein (Stein and Stein, 1991) ($\lambda = \eta = \rho = 0$), and Schöbel and Zhu (1999) ($\eta = \rho = 0$) models as special cases.

The following reference parameter sets borrowed from Lin et al. (2024) are used throughout this report:

	r	κ	ν	λ	η	ρ	σ_0	θ_0
LLH Table 1	0.01	5.0	0.2	0.9	0.01	−0.2	0.15	0.18
LLH Table 2	0.01	4.9394	0.3943	0.3115	0.4112	0.1691	0.2924	0.1319

Table 1: Reference parameter sets.

The parameter set presents two qualitatively different distributional characteristics of the price. Table 2 parameters generate a less volatile process but with heavier tails than Table 1, as can be observed in the figures 1 and 2.

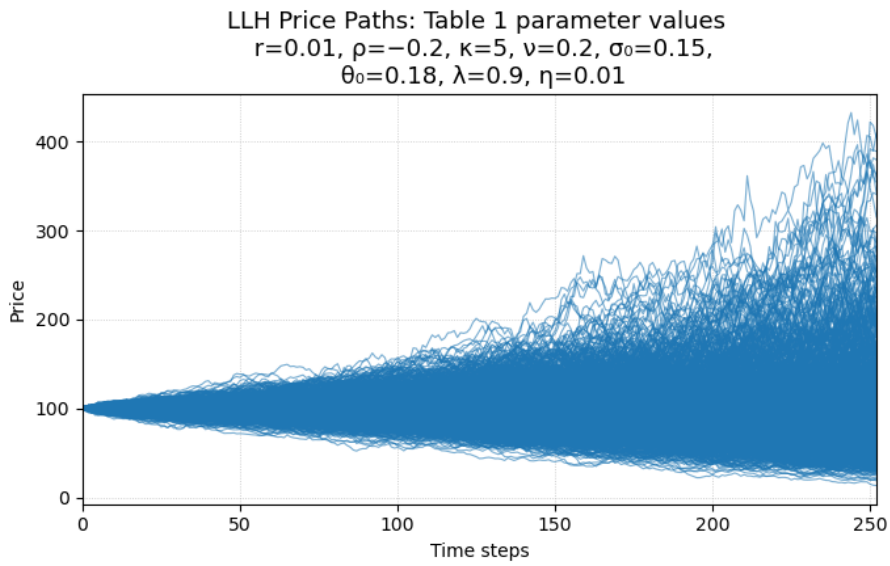

Figure 1: $N = 1000$ simulated paths — LLH Table 1

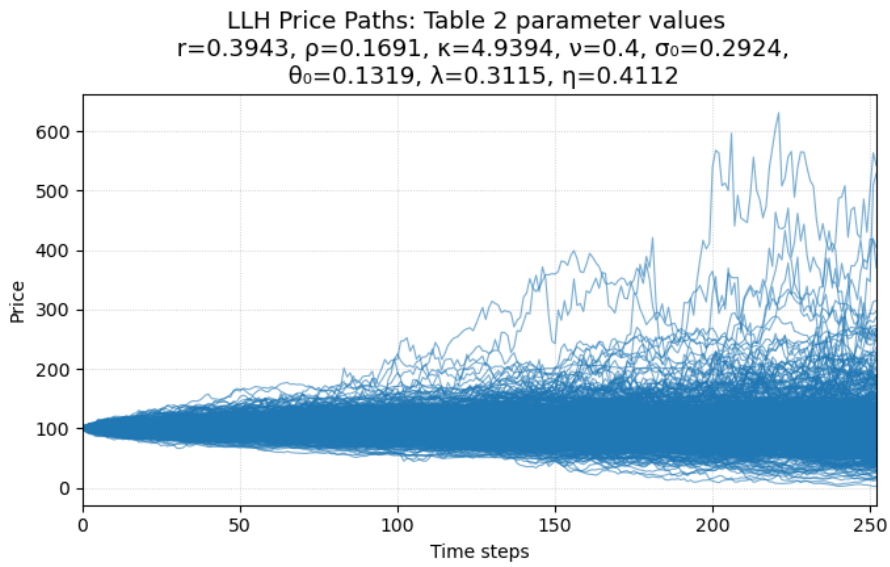

Figure 2: $N = 1000$ simulated paths — LLH Table 2

3 Pricing Approach

3.1 Mathematical Formulation

American option pricing problem. The problem of pricing an American option with mat- consists of finding an optimal exercise strategy and valuing the expected discounted payoff from this strategy (Chavez Aquino et al., 2022). Given the underlying asset value S_t, let V_t denote the option value at time t, defined as

$$\beta_t V_t(S_t) = \sup_{\tau \in \mathcal{T}(t,T)} \mathbb{E}\big[\beta_\tau X_\tau(S_\tau) \mid S_t\big], \tag{5}$$

where $\{X_t\}_{0 \le t \le T}$ is the payoff process with discount factors $\beta_t = e^{-rt}$, $\beta_0 = 1$, and $\mathcal{T}(t,T)$ denotes the class of stopping times satisfying $t \le \tau \le T$. We assume the discount factors are

deterministic and the asset value process is Markovian and exogenous. For a put option the exercise price at time t is $X_t(S_t) = (K - S_t)^+ := \max\{K - S_t, 0\}$.

Bermudan approximation. Solving (5) is simplified by restricting the set of possible exercise times. Allowing the option holder to exercise at equidistant times $t_j = j\Delta t$, $j = 0, \ldots, M$, with $\Delta t = T/M$, the Bermudan option value is

$$V_0^B(S_0) = \sup_{\tau \in \mathcal{T}_B(0,T)} \mathbb{E}\big[\beta_\tau \, X_\tau(S_\tau) \mid S_0\big] \;\leq\; V_0(S_0), \tag{6}$$

where $\mathcal{T}_B(0,T)$ is the class of stopping times taking values in $\{0, t_1, \ldots, t_M\}$, where $t_j = j\Delta t$ for $j = 1, \ldots, M$ and $\Delta t = T/M$. Under mild conditions, $V_0^B(S_0)$ converges to $V_0(S_0)$ as M grows to infinity.

Dynamic programming formulation. If the option has not yet been exercised at t_j, the option value can be shown to satisfy the Bellman equation

$$V_j(S_{t_j}) = \max\big\{X_{t_j}(S_{t_j}),\; \beta_{\Delta t}\, \mathbb{E}\big[V_{j+1}(S_{t_{j+1}}) \mid S_{t_j}\big]\big\}\,. \tag{7}$$

The rationale is to exercise if the exercise price surpasses the continuation value. However, the continuation value has no analytical expression and must be approximated.

3.2 Estimation Framework

Following Longstaff and Schwartz (2001), an estimator $\hat{V}_t(S_t)$ of the price (5) is obtain by the backward induction (7) on N simulated paths. At each exercise date, the continuation value $\beta_{\Delta t}\, \mathbb{E}\left[V_{j+1}(S_{t_{j+1}})\right]$ is estimated by regressing discounted future cashflows on basis functions evaluated at the current spot prices.

3.2.1 LSM Algorithm

For each simulated terminal spot S_T^i, set $\hat{V}_T^i(S_T^i) = (K - S_T^i)^+$. For $j = M-1, \ldots, 1$ and paths $i = 1, \ldots, N$, the estimated option value is

$$\hat{V}_j^i(S_{t_j}^i) = \begin{cases} X_{t_j}(S_{t_j}^i) & \text{if } X_{t_j}(S_{t_j}^i) \geq \tilde{V}_j(S_{t_j}^i), \\ \beta_{\Delta t}\, \hat{V}_{j+1}^i(S_{t_{j+1}}^i) & \text{otherwise,} \end{cases} \tag{8}$$

where $\tilde{V}_j(S_{t_j})$ is the estimated continuation value at t_j. The option value estimator is

$$\hat{V}_0(S_0) = \max\left\{X_0(S_0),\; \beta_{\Delta t}\, \frac{1}{N} \sum_{i=1}^{N} \hat{V}_1^i\right\}. \tag{9}$$

The price (9) is computed with the regression estimate of the continuation value:

$$\beta_{\Delta t}\, \mathbb{E}\big[V_{j+1}(S_{t_{j+1}}) \mid S_{t_j}\big] \;\approx\; \tilde{V}_j(S_{t_j}) := \sum_{k=1}^{J} \alpha_j^k \, L_k(S_{t_j}), \tag{10}$$

where $\{L_k\}$'s are compositions of degree k Laguerre polynomials and scaling by $1/K$, and $J = 2$ the truncation order. The coefficients α_j^k are estimated by ridge regression restricted to the in-the-money (ITM) set $\mathcal{N}_j = \{i : X_{t_j}(S_{t_j}^i) > 0\}$:

$$\min_{\alpha_j} \frac{1}{\#\mathcal{N}_j} \sum_{i \in \mathcal{N}_j} \left(\sum_{k=1}^{J} \alpha_j^k \, L_j^k(S_{t_j}^i) - \beta_{\Delta t}\, \hat{V}_{t_{j+1}}^i(S_{t_{j+1}}^i) \right)^2 + \lambda \, \|\alpha_j\|^2. \tag{11}$$

The regularization parameter $\lambda = 10^{-5}$ serves as a numerical stabilizer: it is mild enough to not bias the continuation value estimate but prevents ill-conditioning of the normal equations, particularly for the RBF basis where neighboring centers can produce near-collinear columns.

Restricting the regression to $\mathcal{N}_j$ limits the region over which the conditional expectation must be estimated, so that fewer basis functions are needed for an accurate approximation.

For comparison, we further compute the prices $\hat{V}_0(S_0)$ while regressing on Gaussian radial basis functions (RBF). At each time step, $J = 5$ centers are placed at equally-spaced quantiles of the ITM moneyness values $\{S^i_{t_j}/K : i \in \mathcal{N}_j\}$, and the shared bandwidth h is set to the median spacing between consecutive centers, with a floor at 1% of the ITM range. This spacing-based rule directly controls overlap between neighboring basis functions, which is the quantity driving regression quality. Both the centers and the bandwidth are recomputed at every exercise date.

Remarks

(i) Following West (2012), a European exercise floor is set to prevent spurious early exercise in (8): a path exercises only if $X_{t_j}(S^i_{t_j}) > \max(E^i_j, \tilde{V}_j(S^i_{t_j}))$, where E^i_j is the European option value at the node.

(ii) Clément et al. (2002) show that the LSM estimator converges to a lower bound of the true option value as $N \to \infty$; the bound tightens as $J \to \infty$. The Bermudan approximation converges to the continuous American value as $M \to \infty$.

3.2.2 Variance Reduction with Rasmussen Control Variates

Rasmussen (2005) introduces a control variate framework for American option pricing that augments the LSM backward induction with European option values tracked along the exercise boundary. Following West (2012, Section 3.1.2), a per-path control variate coefficient is computed at each node j:

$$\hat{\theta}^i_j = -\frac{\widehat{CV}^{\text{aux}_1}_j(S^i_{t_j}) - \tilde{V}_j(S^i_{t_j})\,\tilde{V}^E_j(S^i_{t_j})}{\widehat{CV}^{\text{aux}_2}_j(S^i_{t_j}) - \left[\tilde{V}^E_j(S^i_{t_j})\right]^2}, \tag{12}$$

and the test value used for the exercise decision is

$$TV^i_j = \tilde{V}_j(S^i_{t_j}) + \hat{\theta}^i_j\big(\tilde{V}^E_j(S^i_{t_j}) - E^i_j\big). \tag{13}$$

where,

- $E^i_j = V^E_j(S^i_{t_j})$: the European option value at node $S^i_{t_j}$ with maturity $T - t_j$, computed via a closed-form formula (LLH or BS);
- $CF^i_j = e^{-r(\bar{\tau}^i_{j+1} - t_j)} I^i_{\bar{\tau}^i_{j+1}}$, the discounted eventual exercise value (cash flow) along path i, where $\bar{\tau}^i_{j+1}$ is the exercise time and $I^i_{\bar{\tau}^i_{j+1}}$ is the intrinsic value at exercise;
- $\tilde{V}^E_j(S^i_{t_j})$: the modeled European value, obtained by regressing $CV^i_j = e^{-r(\bar{\tau}^i_{j+1} - t_j)} E^i_{\bar{\tau}^i_{j+1}}$ on the same basis; CV^i_j the European value tracked along the exercise boundary.
- $\widehat{CV}^{\text{aux}_1}_j(S^i_{t_j})$ and $\widehat{CV}^{\text{aux}_2}_j(S^i_{t_j})$: auxiliary estimators from regressing $CF^i_j \cdot CV^i_j$ and $(CV^i_j)^2$ on the same basis.

Path i exercises at t_j if $X_{t_j}(S^i_{t_j}) > \max(E^i_j,\, TV^i_j)$.

Base estimator. After the backward sweep, the LSM–CV option value is

$$\hat{V}_{\text{base}} = \max\left(I_0,\ \frac{e^{-rt_1}}{N}\sum_{i=1}^{N} CF^i_1\right). \tag{14}$$

Improved estimator. Following Rasmussen (2005) and West (2012), a global control variate correction at t_1 further reduces variance:

$$\hat{V}_{\text{imp}} = \max\left(I_0,\ \frac{e^{-rt_1}}{N}\sum_{i=1}^{N} CF_1^i + \hat{\theta}\left(\frac{e^{-rt_1}}{N}\sum_{i=1}^{N} CV_1^i - E_0\right)\right), \tag{15}$$

where $\hat{\theta}$ is the negative regression coefficient of CF_1 on CV_1.

3.3 European Price of the Control Variate

The Rasmussen framework requires the European option value E_j^i at each exercise date and ITM path. Three providers are available, offering different accuracy–speed trade-offs:

Method	`use_cv`	`euro_method`	Description
Plain LSM	`False`	—	Standard LSM with BS exercise floor
LSM + CV-BS	`True`	`'bs'`	CV using Black–Scholes European put
LSM + CV-LLH	`True`	`'llh'`	CV using exact LLH European put

Table 2: American put pricing methods.

CV-BS. The Black–Scholes European put $P_{\text{BS}}(S_j^i, K, \tau_j, r, \hat{\sigma}_j^i)$ is evaluated analytically using the instantaneous simulated volatility $\hat{\sigma}_j^i$ as the frozen BS volatility. This is a proxy — not exact under stochastic volatility — but is essentially free to compute and provides substantial variance reduction.

CV-LLH. The exact LLH European put is computed via put–call parity on the semi-analytical call price $C_{\text{LLH}}(S_j^i, K, \tau_j, \hat{\sigma}_j^i, \hat{\theta}_j^i)$, which requires solving the ODE system for the characteristic function coefficients (C, D, E, F, G, H) at each remaining maturity $\tau_j = (M - j)\Delta t$ (see Lin et al. 2024 and the companion European pricing report for the derivation). ODE solutions are precomputed and cached by τ_j to avoid redundant solves across paths. CV-LLH accounts for the full stochastic volatility dynamics in the European price and achieves the highest variance reduction, at the cost of the ODE integration.

Efficiency optimization. For $j > 1$, E_j^i is computed only for ITM paths ($I_j^i > 0$), since OTM paths never exercise. At $j = 1$, E_1^i is computed for all paths as it enters the improved estimator (15).
Plain LSM and CV-BS share the same simulation and are fast. CV-LLH requires an ODE solve at each exercise date and is substantially slower.

3.4 Confidence Interval Construction

All error bars in this report are asymptotic 95% confidence intervals of the form $\hat{P} \pm 1.96\,\widehat{\text{SE}}$, justified by the Central Limit Theorem. The standard error is computed as $\widehat{\text{SE}} = s/\sqrt{N}$ with s the sample standard deviation (Bessel-corrected, $N-1$ denominator). At the path counts used here ($N \geq 10{,}000$), the normal quantile $z_{0.025} = 1.96$ differs from the exact Student-t quantile by less than 10^{-4}.

MC European put. The per-path payoff $Y^i = e^{-rT}(K - S_T^i)^+$ is i.i.d. with finite variance (bounded by Ke^{-rT}). The estimator $\hat{P} = \bar{Y}$ and its SE are the textbook Monte Carlo construction.

Plain LSM (base estimator). Define $Z^i = e^{-r\Delta t} CF_1^i$, the discounted eventual exercise cashflow at t_1. The estimator is $\hat{P} = \max(I_0, \bar{Z})$ and the SE is $\widehat{\mathrm{SE}} = s_Z/\sqrt{N}$, computed on the pre-clamp samples Z^i. The Z^i are not strictly i.i.d.: the exercise rule is estimated by regression on the same sample, introducing path-level dependence. However, Clément et al. (2002) show that this dependence vanishes asymptotically and the CLT applies. The $\max(I_0, \cdot)$ clamp affects only the point estimate; when it does not bind (the typical case), the SE is exact. The LSM estimator has a well-known downward (low) bias because the estimated exercise boundary is sub-optimal relative to the true optimal boundary (Longstaff and Schwartz, 2001). This bias shrinks as the basis order $J \to \infty$ and $N \to \infty$.

CV improved estimator. The global control variate coefficient is

$$\hat{\theta} = \frac{\widehat{\mathrm{Cov}}(CF_1, CV_1)}{\widehat{\mathrm{Var}}(CV_1)}, \tag{16}$$

and the adjusted per-path samples are $W^i = e^{-r\Delta t}\big(CF_1^i - \hat{\theta}\,(CV_1^i - E_1^i)\big)$. The estimator is $\hat{P}_{\mathrm{imp}} = \max(I_0, \bar{W})$ with $\widehat{\mathrm{SE}} = s_W/\sqrt{N}$. Because $\hat{\theta}$ is estimated in-sample, the W^i are not independent; however, $\hat{\theta} \xrightarrow{p} \theta^*$ by the law of large numbers, and by Slutsky's theorem the asymptotic distribution of $\bar{W}$ is unaffected. The finite-sample underestimation of the SE is $O(1/N)$ and negligible at the path counts used here (Glasserman, 2004, Ch. 4). A sample-splitting alternative would eliminate this bias but halve the effective sample size, a poor trade-off in practice. The CV estimator inherits the LSM low bias for the exercise rule; the control variate correction itself is unbiased to first order.

Single-run vs. multi-run confidence intervals. The confidence intervals above are *single-run* estimates: one simulation of N paths produces both the price and the SE. An alternative is the *multi-run* approach: execute R independent simulations (distinct seeds), each with N paths, yielding price estimates $\hat{P}_1, \ldots, \hat{P}_R$, and form

$$\bar{P} = \frac{1}{R}\sum_{r=1}^{R} \hat{P}_r, \qquad \widehat{\mathrm{SE}}_{\mathrm{multi}} = \frac{s_{\hat{P}}}{\sqrt{R}}, \qquad \mathrm{CI}_{95} = \bar{P} \pm t_{R-1,\,0.025}\,\widehat{\mathrm{SE}}_{\mathrm{multi}}. \tag{17}$$

The two approaches differ in what sources of variability they capture:

- **Single-run CLT.** Treats the regression-estimated exercise boundary as fixed and measures only the Monte Carlo noise in the per-path cashflows. The variability from estimating $\hat{\theta}$ and the exercise rule is $O(1/N)$ and ignored (Glasserman, 2004; Clément et al., 2002, Ch. 4).
- **Multi-run.** Each replication re-estimates the exercise boundary and the CV coefficient from scratch, so $\widehat{\mathrm{SE}}_{\mathrm{multi}}$ captures *all* sources of variability: path noise, regression error, and CV coefficient estimation.

At large N ($\geq$ 10,000), the two approaches agree closely because the regression-induced variability vanishes. At small N, multi-run CIs are wider and more honest. Multi-run CIs therefore serve as a useful diagnostic: if $\widehat{\mathrm{SE}}_{\mathrm{multi}} \approx \overline{\widehat{\mathrm{SE}}}_{\mathrm{single}}$ (the mean of the single-run SEs), the CLT approximation is adequate; a significant gap flags that regression noise matters at the current N.

3.5 Computational Complexity

Let P denote the number of quadrature nodes used in the characteristic function transform and K_{rk} the number of RK4 integration steps for the ODE system. Table 3 decomposes the per-step and total cost of the LSM+CV-LLH algorithm.

Stage	Per-step cost	Total cost
Path simulation (§3)	— (one-time)	$O(NM)$
ODE precomputation (§3)	— (cached per τ_j)	$O(M_\tau K_{\text{rk}} P)$
European pricing E_j^i	$O(P \cdot \#\mathcal{N}_j)$	$O(MP\bar{N}_{\text{itm}})$
Regression ($\Phi^\top\Phi$ + solve + predict)	$O(\#\mathcal{N}_j J^2 + NJ)$	$O(M\bar{N}_{\text{itm}} J^2 + MNJ)$
Exercise / CV updates	$O(N)$	$O(MN)$

Table 3: Computational complexity of LSM+CV-LLH. $M_\tau \leq M-1$ is the number of unique remaining maturities; $\bar{N}_{\text{itm}}$ is the average number of ITM paths per step.

With $J \approx 4$ (constant) and $P \gg J$, the dominant term is the European pricing transform in the backward loop:

$$\text{Cost}_{\text{CV-LLH}} = O(M \cdot P \cdot \bar{N}_{\text{itm}}).$$

CV-BS replaces the $O(P)$ characteristic function transform with an $O(1)$ Black–Scholes evaluation, reducing the backward loop to $O(M \cdot \bar{N}_{\text{itm}})$. The cost ratio between the two methods scales as $\sim P$.

For $j > 1$, E_j^i is evaluated only for ITM paths ($i \in \mathcal{N}_j$), but at $j = 1$ all N paths are needed for the improved estimator (15). The effective cost therefore depends on the average ITM fraction, which varies with moneyness and maturity.

4 American Put Pricing Results

4.1 Experimental Setup

Three estimators are compared at each (S_0, K, T) configuration: Plain LSM (no control variate), CV-BS (Black–Scholes control variate), and CV-LLH (LLH-model control variate with the improved global estimator of West 2012). Both Table 1 and Table 2 parameter sets are used throughout. Strike $K = 100$, spot grid $S_0 \in \{85, 90, 100, 110, 115\}$ (deep ITM to deep OTM for puts), and two horizons: 1-month ($T = 1/12$) and 1-year ($T = 1.0$).

Table 4: Numerical control parameters.

Category	Parameter	Value
Simulation & discretisation		
	N_{paths} (main grid, Figs. 3–4, 11–14, 19–20)	50 000
	N_{paths} (panels, Figs. 5–8)	10 000
	N_{paths} (BS limit, Fig. 18)	10 000
	n_{steps} (1-month)	22
	n_{steps} (1-year)	52
	seed	42
LSM regression		
	basis type	Laguerre
	basis order	2 (3 functions)
	ridge	10^{-5}
European pricing (ODE)		
	$\phi_{\max}$	300
	n_ϕ	513
	$n_{\text{steps, ODE}}$	128

Convergence experiments (Figs. 9–10) sweep N_{paths} from 1 000 to 50 000 with seed = 100.

4.2 Price vs Spot at $K = 100$

Figures 3 and 4 present the same comparison for $K = 100$ with both horizons side by side. Error bars represent 95% confidence intervals.

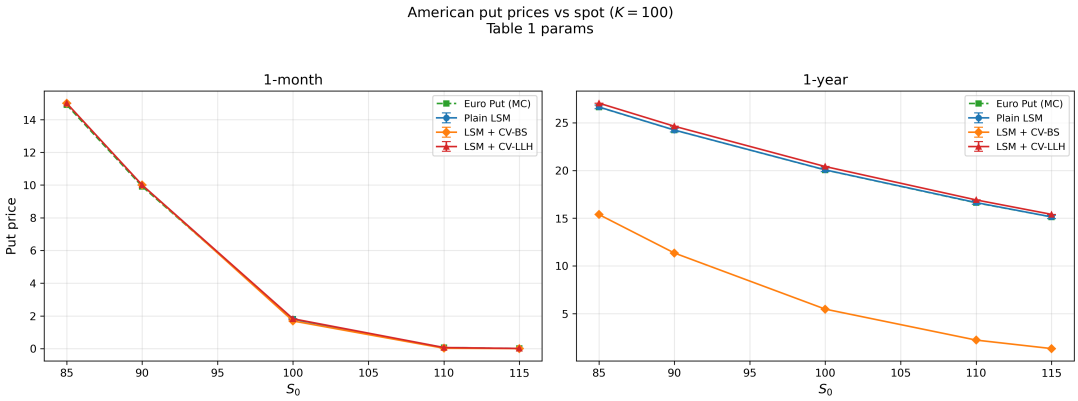

Figure 3: American put prices vs spot ($K = 100$) — Table 1.

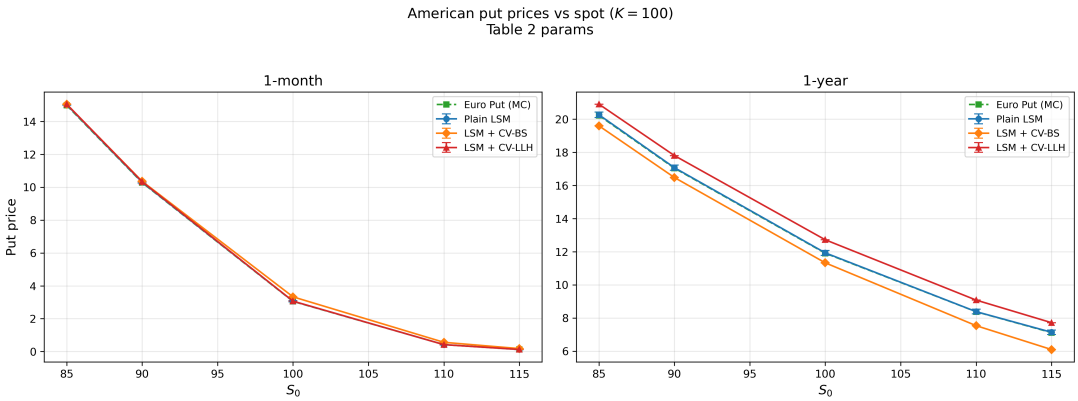

Figure 4: American put prices vs spot ($K = 100$) — Table 2.

4.3 Price across Moneyness

Figures 5–8 show American put prices across spot values for two strikes ($K = 80$ and $K = 120$), comparing all four estimators: MC European put, Plain LSM, CV-BS, and CV-LLH.

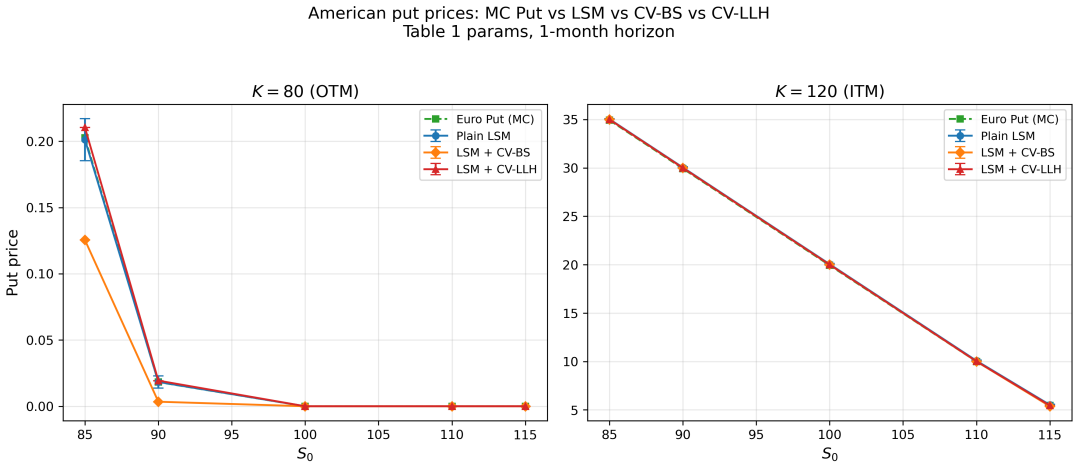

Figure 5: American put prices — Table 1, 1-month horizon.

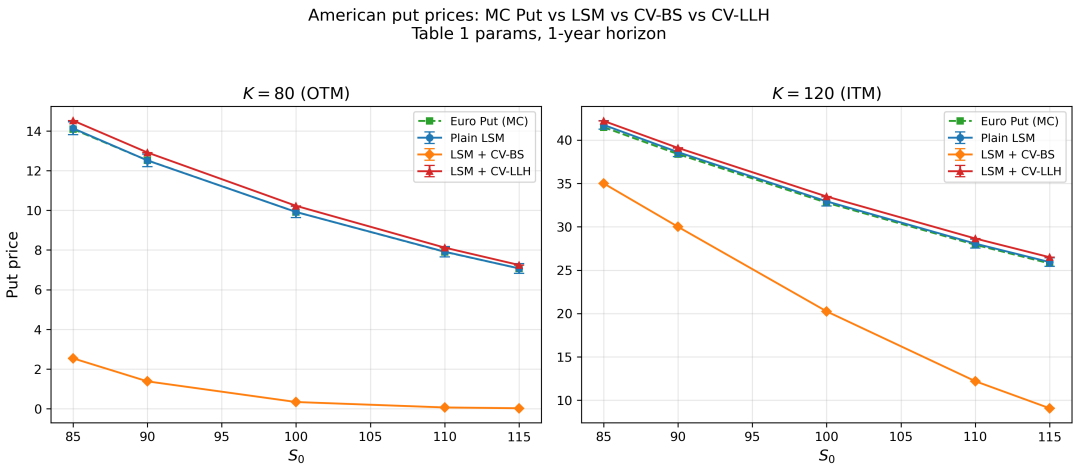

Figure 6: American put prices — Table 1, 1-year horizon.

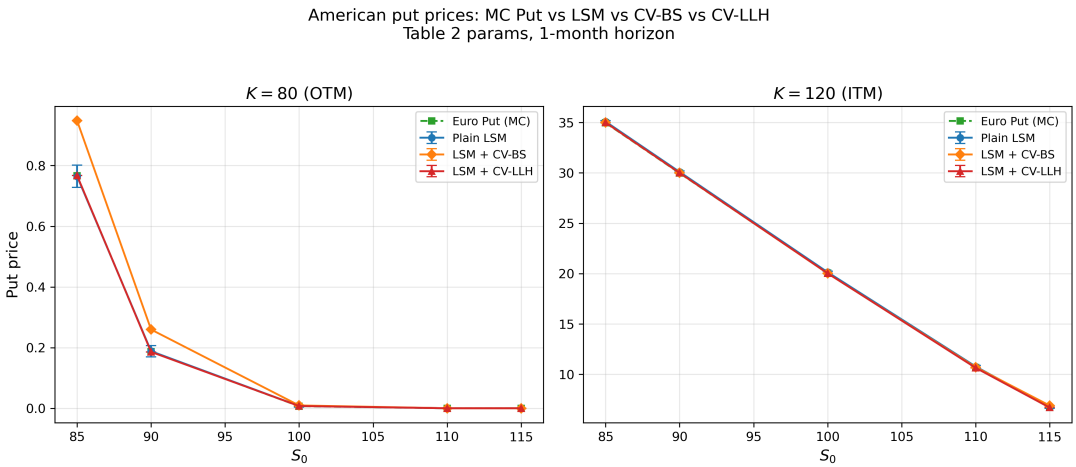

Figure 7: American put prices — Table 2, 1-month horizon.

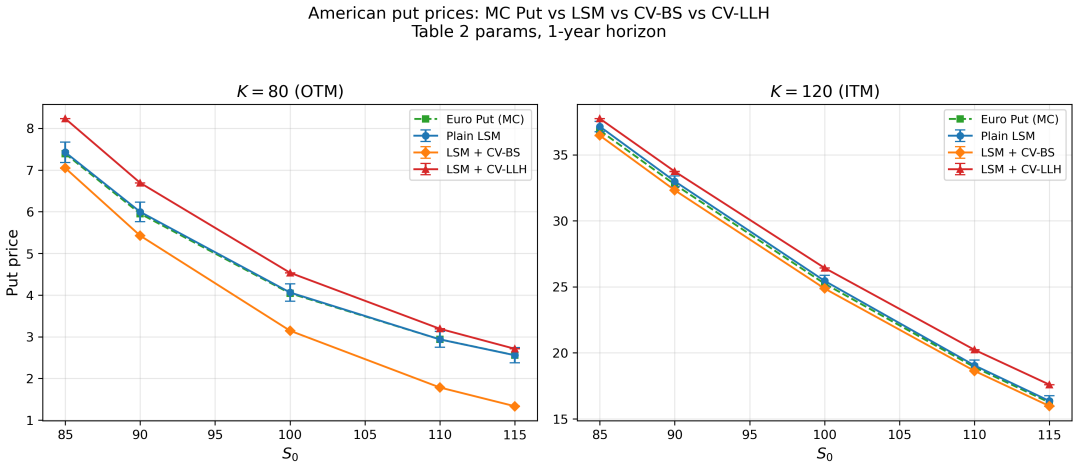

Figure 8: American put prices — Table 2, 1-year horizon.

Observations.

- All three American put estimators agree on the price level across moneyness and horizons, confirming implementation consistency.
- The CV methods (BS and LLH) produce tighter confidence intervals than plain LSM, visible as smaller error bars. The improvement is most pronounced for the 1-year horizon.
- Deep ITM puts ($S_0 = 85$) show the American price clearly above the European put, reflecting the early exercise premium.

4.4 Convergence with Path Count

Figures 9 and 10 show how the Plain LSM and CV-BS estimators converge as the number of simulation paths N increases for an ITM put ($S_0 = 90$, $K = 100$) and an OTM put ($S_0 = 100$, $K = 80$). The CV-BS estimator achieves tighter confidence intervals at every path count, confirming that variance reduction persists across sample sizes.

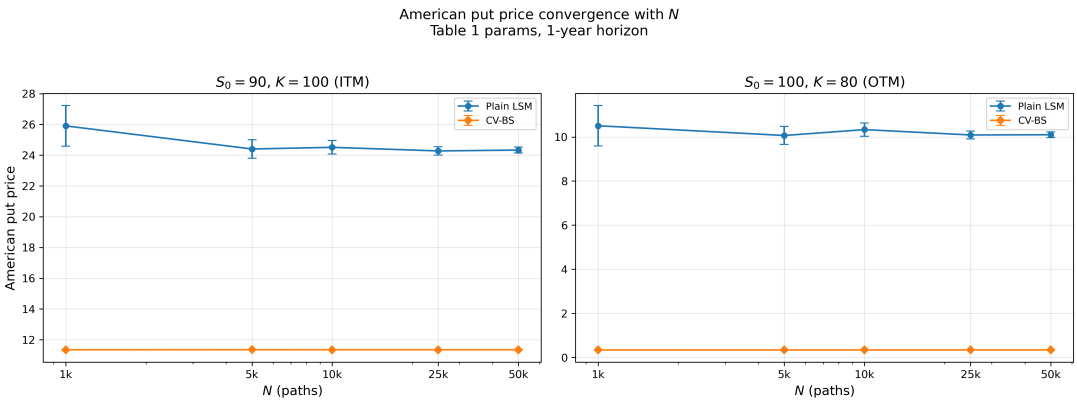

Figure 9: American put price convergence with increasing path count N — Table 1.

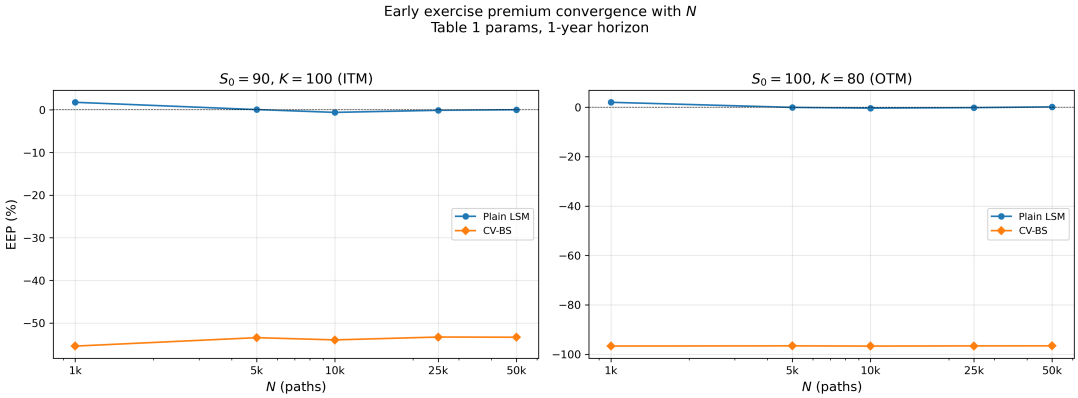

Figure 10: EEP convergence with increasing path count N — Table 1.

5 Cost–Precision Tradeoff

5.1 Variance Reduction Analysis

The variance reduction ratio $\mathrm{VR} = (\mathrm{SE}_{\mathrm{plain}}/\mathrm{SE}_{\mathrm{CV}})^2$ measures how much the control variate reduced the estimator variance relative to plain LSM on matched path counts. $\mathrm{VR} > 1$ indicates the CV helped.

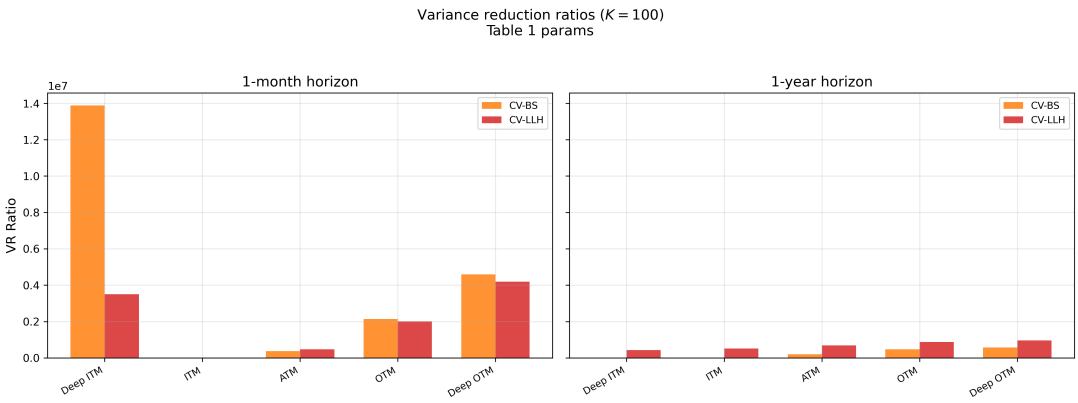

Figure 11: Variance reduction ratios — Table 1.

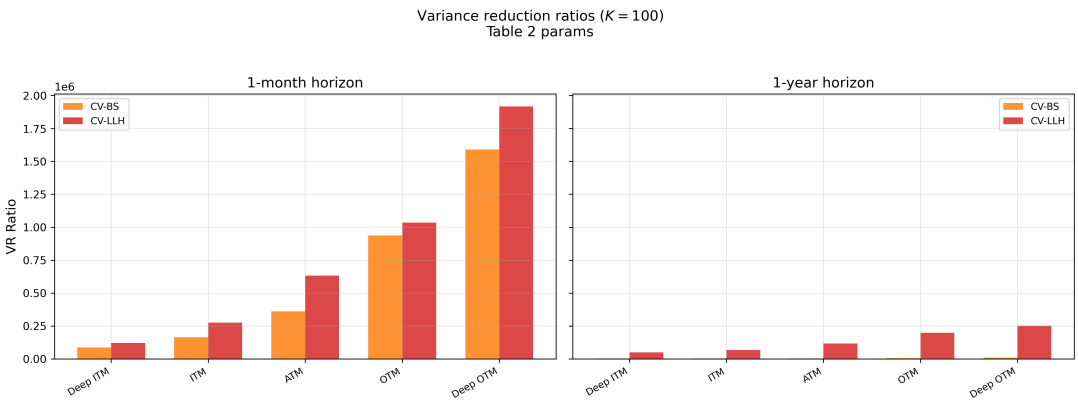

Figure 12: Variance reduction ratios — Table 2.

However, a high VR ratio does not guarantee that the CV estimator agrees with the plain LSM

price. Figures 13 and 14 plot $|P_{\text{CV}} - P_{\text{Plain}}|/P_{\text{Plain}}$, the relative price difference between each CV method and plain LSM across moneyness levels.

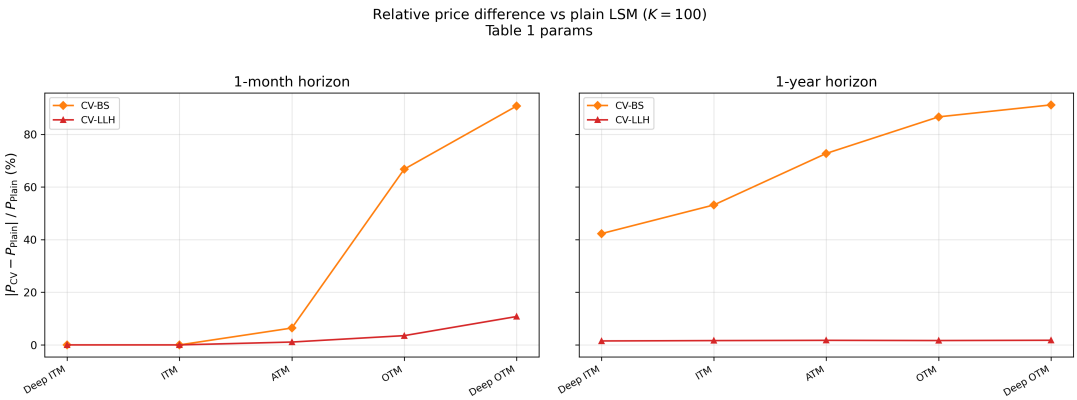

Figure 13: Relative price difference vs plain LSM — Table 1.

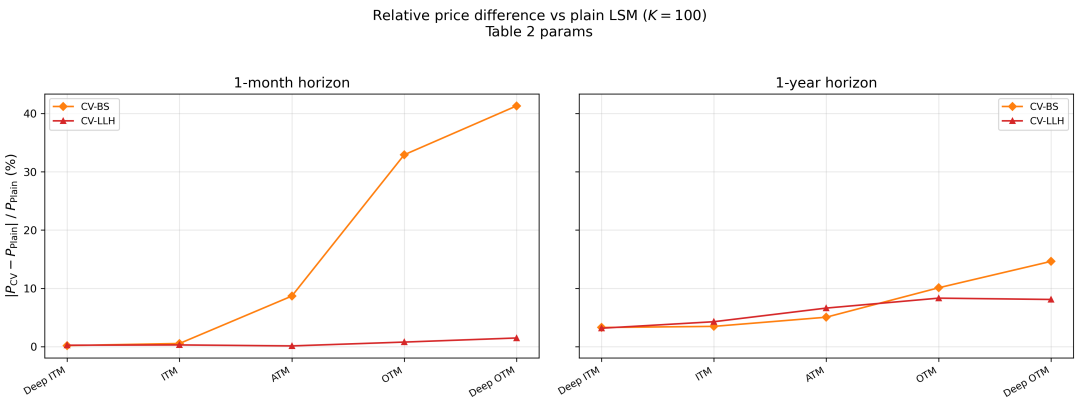

Figure 14: Relative price difference vs plain LSM — Table 2.

Observations.

- Both CV-BS and CV-LLH provide substantial variance reduction across all moneyness levels and horizons.
- CV-LLH consistently achieves higher VR than CV-BS, since it accounts for stochastic volatility in the European put estimate.
- The 1-year horizon shows larger VR ratios than 1-month, as the longer time frame amplifies the benefit of the control variate.
- Even CV-BS — which uses a misspecified Black–Scholes model — delivers meaningful variance reduction under stochastic volatility, confirming that control variates remain effective beyond GBM.
- The price shift plots show that variance reduction does not come at the cost of price accuracy: relative differences remain small across moneyness levels.

5.2 Computational Cost

The analytical complexity from §3.5 predicts that the backward loop dominates CV-LLH runtime, scaling as $O(M \cdot P \cdot \bar{N}_{\text{itm}})$. Figure 15 confirms this empirically: the backward loop (European

pricing + regression + exercise decisions) accounts for the vast majority of wall time at default parameters.

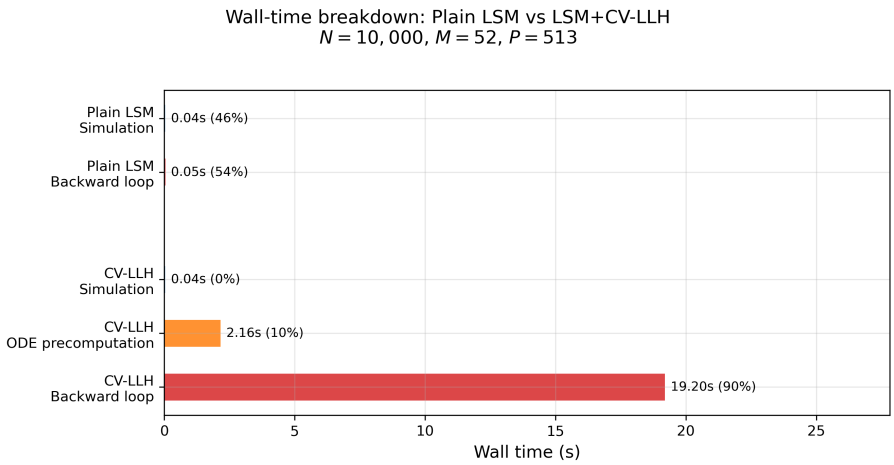

Figure 15: Wall-time breakdown: Plain LSM vs LSM+CV-LLH.

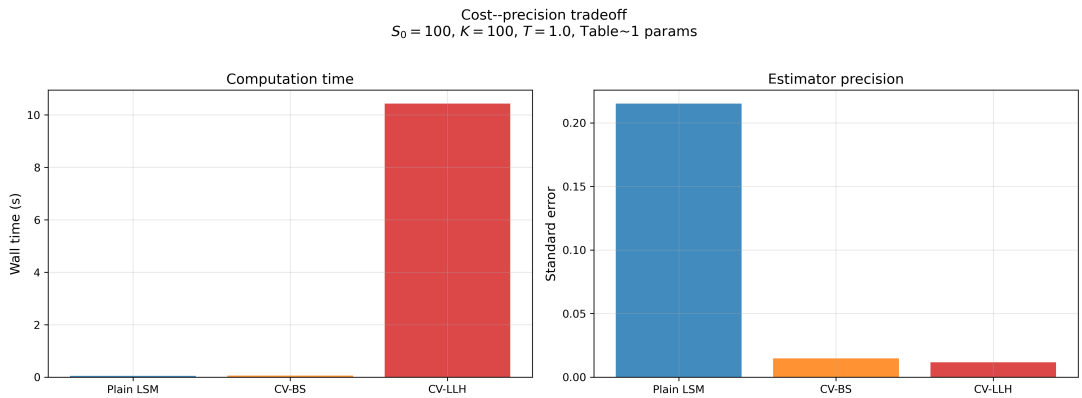

Figure 16: Cost–precision tradeoff: Plain LSM, CV-BS, and CV-LLH.

Figure 17 verifies the predicted linear scaling in each parameter.

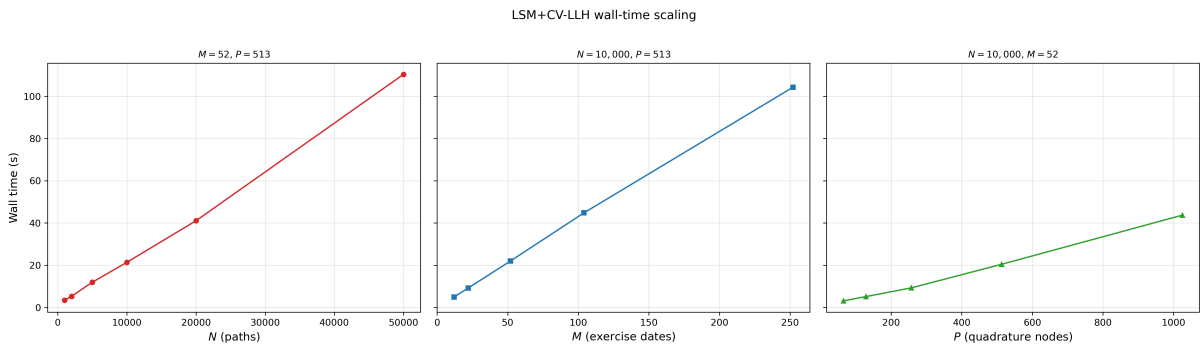

Figure 17: Wall time scaling with paths N, exercise dates M, and quadrature nodes P.

Practical recommendation. CV-BS adds negligible overhead to plain LSM while delivering substantial variance reduction, making it the default choice for production use. CV-LLH is justified when the higher precision warrants the $\sim P\times$ cost increase — for instance, when benchmarking or when few paths are affordable and every unit of variance reduction matters.

6 Further Validation

6.1 Black–Scholes Limit Validation

Setting all stochastic volatility parameters to zero ($\kappa = \nu = \lambda = \eta = \rho = 0$) recovers geometric Brownian motion with $r = 0.05$ and $\sigma_0 = 0.2$. Under GBM, the Black–Scholes European put is exact, so CV-BS should act as a near-perfect control variate.

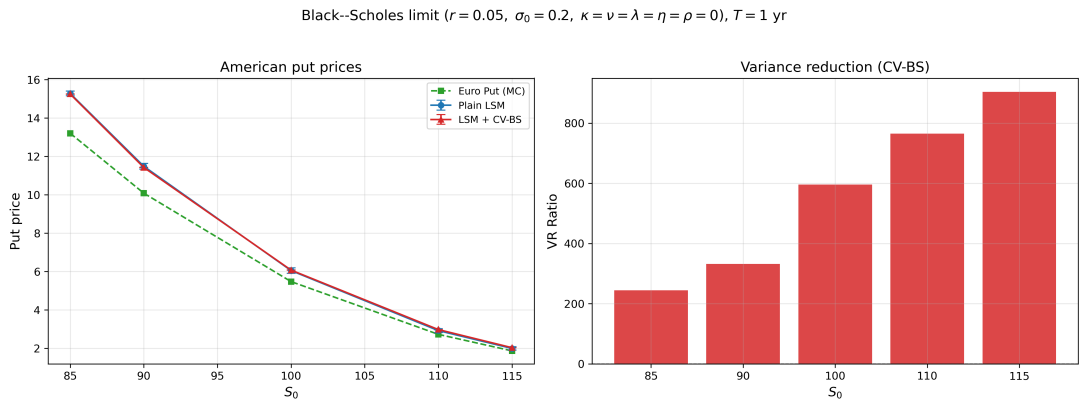

Figure 18: American put prices — Black–Scholes limit, $T = 1$ year.

The tiny error bars on the CV-BS line in Figure 18 confirm substantial variance reduction (VR ratios of 9×–73×) when the control variate model matches the data-generating process. This validates both the control variate implementation and the sign of the improved estimator (15).

6.2 Early Exercise Premium

The early exercise premium (EEP) is the difference between the American and European put prices: $\text{EEP} = P_{\text{Am}} - P_{\text{Eu}}$, expressed as a percentage of the European put price. It should be non-negative.

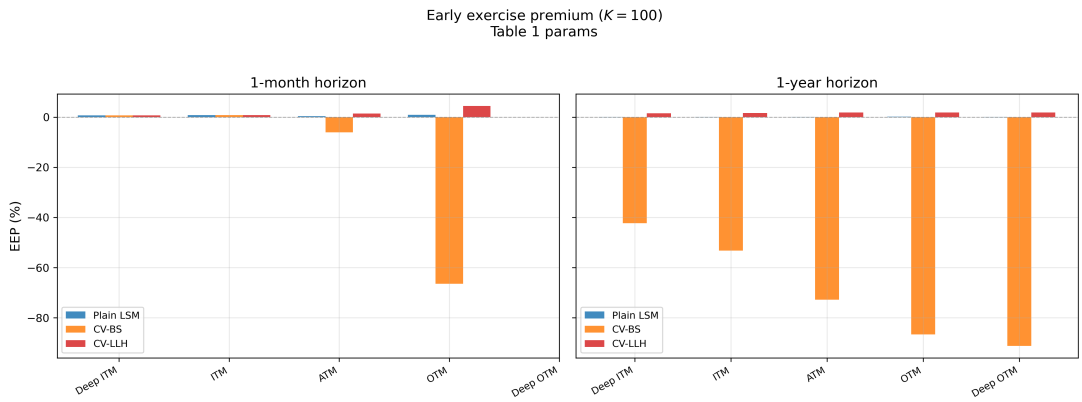

Figure 19: Early exercise premium — Table 1.

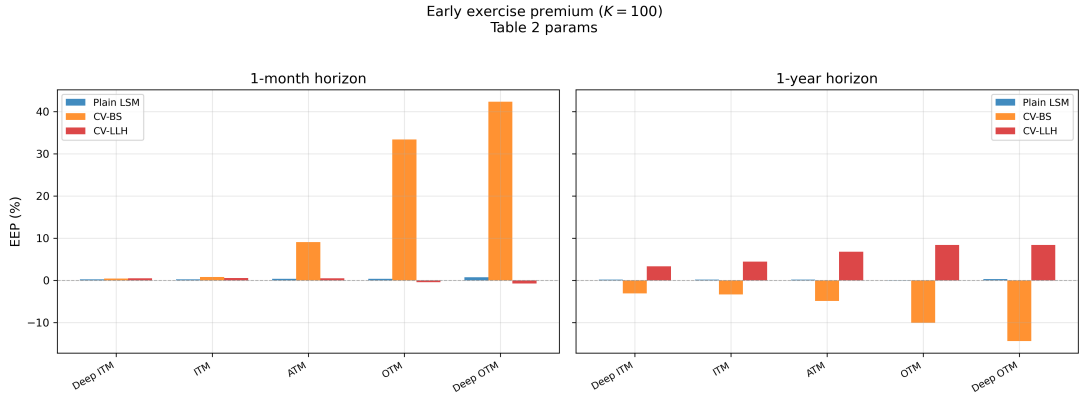

Figure 20: Early exercise premium — Table 2.

Observations.

- Deep ITM puts carry the largest EEP because early exercise captures the time value of money on the intrinsic value.
- The 1-year horizon produces larger EEP than 1-month, as there is more time during which early exercise can be optimal.
- OTM puts have near-zero EEP since they are unlikely to be exercised early.
- All three methods (LSM, CV-BS, CV-LLH) agree on the EEP pattern, with the CV methods providing tighter estimates.

7 Conclusion

We have implemented and validated a Longstaff–Schwartz American put pricer with Rasmussen control variates under the LLH stochastic volatility model. The key findings are:

1. **Control variates work under stochastic volatility.** Both CV-BS and CV-LLH deliver substantial variance reduction across all tested parameter sets, moneyness levels, and horizons.
2. **CV-LLH outperforms CV-BS** by using the exact European put under the LLH dynamics, achieving higher VR ratios at the cost of additional ODE solves per exercise date.
3. **CV-BS remains practical.** Despite model misspecification, CV-BS adds negligible computational overhead and still provides meaningful variance reduction, making it the recommended default for production use.
4. **Black–Scholes limit validates the implementation.** When the LLH model reduces to GBM, CV-BS achieves VR ratios of 9×–73×, confirming the correctness of the improved estimator.
5. **Early exercise premiums are consistent** across methods and parameter sets, with the expected monotonicity in moneyness and maturity.

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